Annual Evaluation March 2026 Class 9 - BIOLOGY

Qn	**Answer Key (English medium)** Rasheed Odakkal	
1	a). Lenticels. **2**. b). p-z, q-x, r-y	
3	c). Both A and R are correct; R is the correct explanation of A. **4**. b). ii – iv – iii – i.	**4x1**
5	a). Bacteria, Archaea. b). The structural differences, diverse adaptations for surviving the environmental changes and differences in the genetic materials.	
6	a). Osmosis. b). No, the entry of oxygen into cells occur through diffusion.	
7	Yes. Balnced diet, adequate water, reduce salt and fat, rest, exercice, checkup, mental support ….	
8	No. Phylum chordata includes organisms with notochord or vertebral column. In subphylumVertebrata, the notochord is seen only during the embryonic stage, and as the embryo grows, it ransforms into the vertebral column.	
9	a).X- Flexor muscle, Y- Extensor muscle. b). When folding hands, flexor muscle contracts, while the extensor muscle stretches or relaxes.	
10	**A** a). P- Gastric juice, Q- Saliva. b). R (bile) converts fat into small particles and regulates pH. **B** In amoeba, pseudopodia are the means of ingestion, while in hydra, these are tentacles. Digestion in amoeba is intracellular, where as in hydra it is extracelllular and intracellular.	**7x2**
11	This pairing is wrong. The cardiac muscle seen here can be seen only in the heart. Stomach wall is made up of smooth muscles.	
12	The substance abuse affects fertility and reproductive health. They cause hormonal imbalance, irregular menstruation, ovulation disorders and miscarriage in women. And cause decreased sperm efficiency and motility in men.	
13	**A** a). X- Angiosperms, Y- Gymnosperms. b). Hibiscus, mango tree, coconut tree etc. are angiosperrms. Cycas and pine are gymnosperm. c). Reproductive parts are present in the flowers. Fruits are seen covering the seeds. **B** a). Presence of vertebral column. b). Hair and lactation seen in dog but absent in lizard. c). Lungfish. Because, all others have forelimbs.	
14	a). Moisture can lead to fungal infections. b). It is essential to keep the genital area without moisture. Wear dry, cotton undergarments facilitating air circulation and change them daily.	
15	**A** a). Oxygen is used in Kreb's cycle (second stage). b). Oxygen is transported mainly as oxyhaemoglobin. 70% CO_2 is transported as bicarbonate 23% as carbaminohaemoglobin. **B** a). Hemodialysis. b). 1. Blood with waste materials, is passed to the dialysis unit after adding heparin. 2. Anti heparin is added to the purified blood and is returned to the body.	**6x3**
16	a). Viruses. b). No. Organisms having the ability to reproduce, respond, to grow, and do metabolic activities are included in the process of classification. Viruses cannot multiply without the help of a host cell. They are inactive outside any living cell.	
17	**A** a). Synovial fluid. Cartilage. b). Rheumatoid arthritis. **B** a). Hydrotropism. b). Presence of water. c). After a few days, roots of all pea plants grow around the water containing pot.	
18	**A** a). S - Placenta. b). P - Amniotic fluid. It prevents dehydration of the foetus / protects it from external shock. c). Yes, I agree. Exchange of nutrients from mother blood to foetus through umbilical cord occur by the process of diffusion and facilitated diffusion. **B** a). X- Tubectomy, Y- IUD (Intra Uterine Device). b). Tubectomy is a permanent contraceptive method, IUD is temporary. c). Yes, I agree. Freequent pregnancy and parturition has high risk of health in both the mother and her child.	**1x4** **(40)**